

CHAPTER SIX

PLANAR KINETICS OF A RIGID BODY: FORCE AND ACCELERATION

Learning Objectives

After completing this chapter, students should be able to do the following:

- To determine the mass moment of inertia of a body.
- To develop the planar kinetic equations of motion for a symmetric rigid body.
- To apply equations of mass moment of inertia of a body and planar kinetic equations of motion to bodies undergoing translation, rotation about a fixed axis, and general plane motion.

6.1 Mass Moment of Inertia

It has been established that translational aspects of a motion are governed by the equation $\mathbf{F} = m\mathbf{a}$. Similarly, the rotational aspects, caused by a moment $\mathbf{M}$, are governed by an equation of the form $\mathbf{M} = I\boldsymbol{\alpha}$. The symbol I in this equation is termed the mass moment of inertia. By comparison, the *moment of inertia* is a measure of the resistance of a body to *angular acceleration* ($\mathbf{M} = I\boldsymbol{\alpha}$) in the same way that mass is a measure of the body's resistance to acceleration ($\mathbf{F} = m\mathbf{a}$).

We define the moment of inertia as the integral of the "second moment" about an axis of all the elements of mass dm which compose the body.



Fig. 6.1

For example, the body's moment of inertia about the z axis in Fig. 6.1 is

$$I = \int_m r^2 dm \quad (6.1)$$

Here the "moment arm" r is the perpendicular distance from the z axis to the arbitrary element dm . Since the formulation involves r , the value of I is different for each axis about which it is computed. In the study of planar kinetics, the axis chosen for analysis generally passes through the

body's mass center G and is always perpendicular to the plane of motion. The moment of inertia about this axis will be denoted as I_G .

If the body consists of material having a variable density, $\rho = \rho(x, y, z)$, the elemental mass dm of the body can be expressed in terms of its density and volume as $dm = \rho dV$. Substituting dm into Eq. 6.1, the body's moment of inertia is then computed using volume elements for integration; i.e.

$$I = \int_V r^2 \rho dV \quad (6.2)$$

In the special case of ρ being a *constant*, this term may be factored out of the integral, and the integration is then purely a function of geometry,

$$I = \rho \int_V r^2 dV \quad (6.3)$$

Example 6.1

Determine the moment of inertia of the cylinder shown in Fig. E-6.1a about the z axis. The density of the material, ρ , is constant.

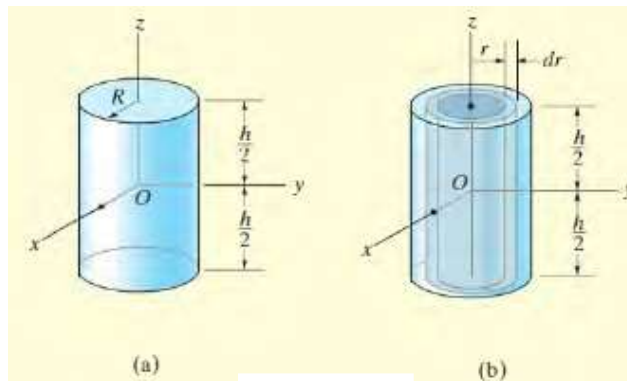


Fig. E-6.1

Solution:

Shell Element: This problem can be solved using the shell element in Fig. E-6.1b and a single integration. The volume of the element is $dV = (2\pi r)(dr)(h)$, so that its mass is $dm = \rho dV = dm = \rho(2\pi hr dr)$. Since the *entire element* lies at the same distance r from the z axis, the moment of inertia of the element is

$$dI_z = r^2 dm = \rho 2\pi h r^3 dr$$

Integrating over the entire region of the cylinder yields

$$I_z = \int_{mi} r^2 dm = \rho 2\pi h \int_0^R r^3 dr = \frac{\rho \pi}{2} R^4 h$$

The mass of the cylinder is

$$m = \int_m dm = \rho 2\pi h \int_0^R r dr = \rho \pi h R^2$$

so that

$$I_z = \frac{1}{2} m R^2$$

Example 6.2

If the density of the material is 515 kg/m^3 , determine the moment of inertia of the solid in Fig. E-6.2a about the y axis. Assume $1 \text{ ft} = 0.3 \text{ m}$.

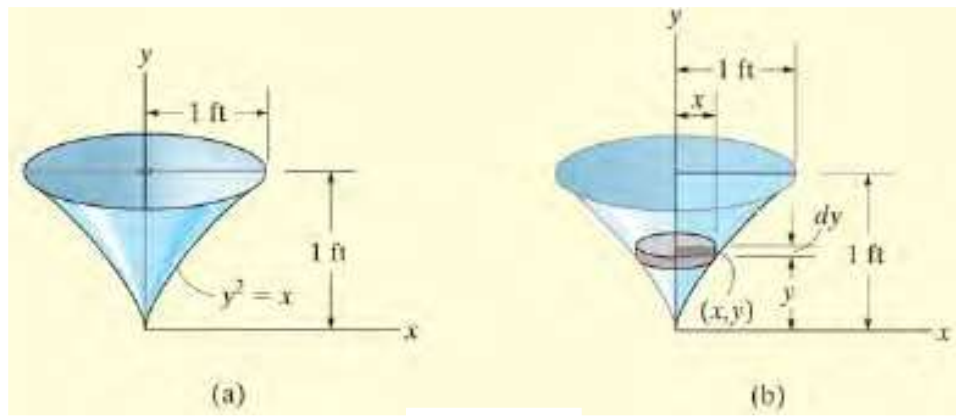


Fig. E-6.2

Solution:

Disk element: The moment of inertia will be found using a disk as shown in Fig. E-6.2b. Here the element intersects the curve at the arbitrary point (x, y) and has a mass

$$dm = \rho dV = \rho(\pi x^2) dy$$

curve at the arbitrary point (x, y) and has a mass

$\therefore$ for the disk element in Fig. 6.2b, we have

$$dI_y = \frac{1}{2} (dm) x^2 = \frac{1}{2} \{ \rho(\pi x^2) dy \} x^2$$

Substituting $x = y^2$, $\rho = 515 \text{ kg/m}^3$, and integrating with respect to y , from $y = 0$ to $y = 0.3 \text{ m}$, yields the moment of inertia for the entire solid.

$$I_y = \frac{\pi(515 \text{ kg/m}^3)}{2} \int_0^{0.3} x^4 dy = \frac{\pi(515)}{2} \int_0^{0.3} y^8 dy = \frac{\pi(515)}{2} \times \left[\frac{y^9}{9} \right]_0^{0.3}$$

$$\therefore I_y = \frac{\pi(515)}{2} \times \left[\frac{(0.3)^9}{9} \right] = 1.77 \times 10^{-3} \text{ kg} \cdot \text{m}^2$$

6.1.1 Parallel-axis theorem

If the moment of inertia of the body about an axis passing through the body's mass center is known, then the moment of inertia about any other parallel axis can be determined by using the parallel-axis theorem.

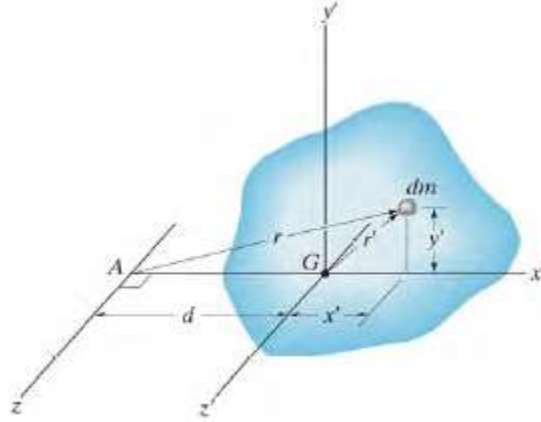


Fig. 6.2

This theorem can be derived by considering the body shown in Fig. 6.2. Here the z' axis passes through the mass center G , whereas the corresponding *parallel* z axis lies at a constant distance d away. Selecting the differential element of mass dm , which is located at point (x', y') , and using the Pythagorean theorem,

$$r^2 = (d + x')^2 + (y')^2$$

Hence, the moment of inertia of the body about the z axis can be expressed as

$$I = \int_m r^2 dm = \int_m [(d + x')^2 + (y')^2] dm$$

$$\therefore I = \int_m [(x')^2 + (y')^2] dm + 2d \int_m x' dm + d^2 \int_m dm$$

Since $(r')^2 = (x')^2 + (y')^2$, the first integral represents I_G . The second integral equals zero, since the z' axis passes through the body's mass center, i.e., $\int x' dm = \bar{x}'m = 0$, since $\bar{x}' = 0$. Finally, the third integral represents the total mass m of the body. Hence, the moment of inertia about the z axis can be written as

$$I = I_G + md^2 \tag{6.3}$$

Where,

I_G = moment of inertia about the z' axis passing through the mass center G

m = mass of the body

d = perpendicular distance between the parallel z and z' axes

6.1.2 Radius of gyration

Occasionally, the moment of inertia of a body about a specified axis is reported in handbooks using the radius of gyration, k . This is a geometrical property which has units of length. When it and the body's mass m are known, the body's moment of inertia is determined from the equation

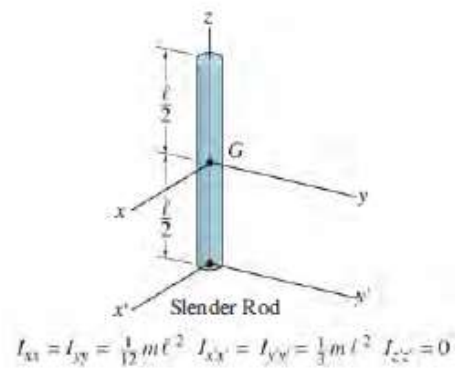
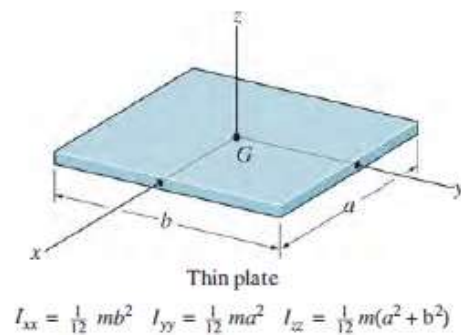
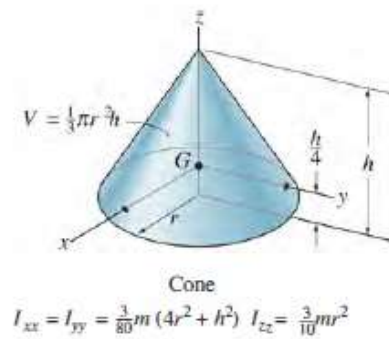
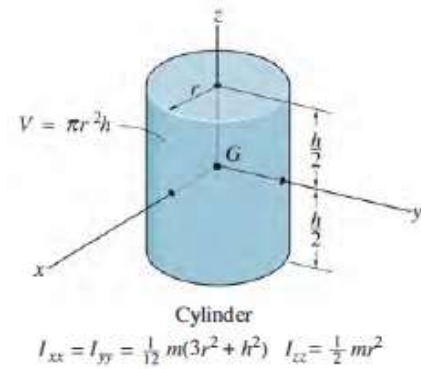
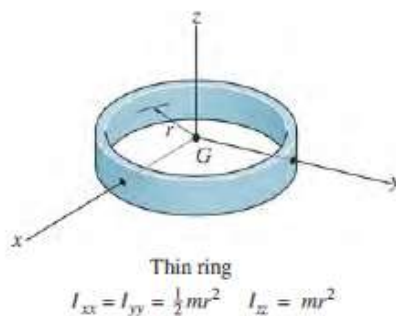
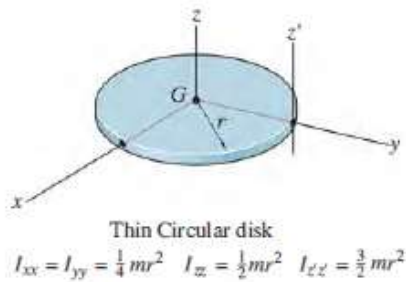
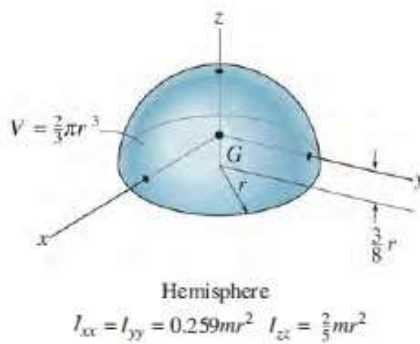
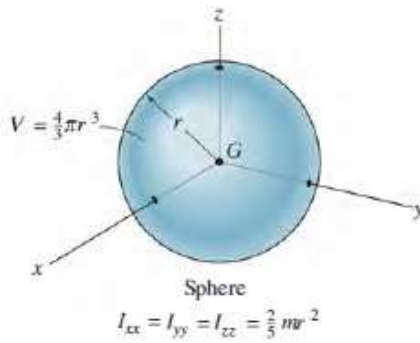
$$I = mk^2 \quad \text{or} \quad k = \sqrt{\frac{I}{m}} \quad (6.4)$$

6.1.3 Composite bodies

If a body consists of a number of simple shapes such as disks, spheres, and rods, the moment of inertia of the body about any axis can be determined by adding algebraically the moments of inertia of all the composite shapes computed about the axis. Algebraic addition is necessary since a composite part must be considered as a negative quantity if it has already been counted as a piece of another part -for example, a "hole" subtracted from a solid plate. The parallel-axis theorem is needed for the calculations if the center of mass of each composite part does not lie on the axis. For the calculation, then, $I = \sum(I_G + md^2)$. Here I_G for each of the composite parts is determined by integration, or for simple shapes, such as rods and disks, it can be found from a table, such as the one given in Table 6.1.

Table 6.1: Center of Gravity and Mass Moment of Inertia of Homogeneous Solids

Center of Gravity and Mass Moment of Inertia of Homogeneous Solids



Example 6.3

If the plate shown in Fig. E-6.3a has a density of 8000 kg/m^3 and a thickness of 10 mm, determine its moment of inertia about an axis directed perpendicular to the page and passing through point O .

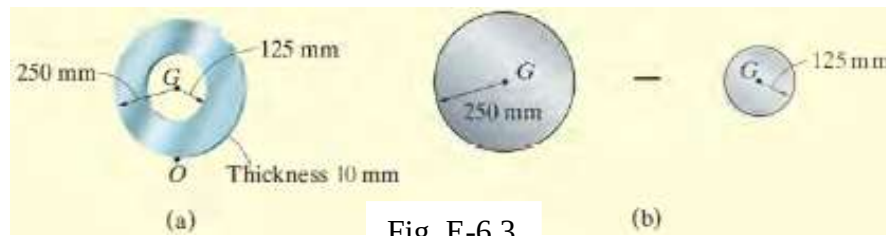


Fig. E-6.3

Solution:

The plate consists of two composite parts, the 250-mm-radius disk minus a 125-mm-radius disk, Fig. E-6.3b. The moment of inertia about O can be determined by computing the moment of inertia of each of these parts about O and then adding the results algebraically (by subtracting the moment of inertia of the 125 mm radius disk from that of the 250 mm radius disk). The calculations are performed by using the parallel-axis theorem in conjunction with the data listed in the table on the inside back cover.

Disk. The moment of inertia of a disk about the centroidal axis perpendicular to the plane of the disk is $I_G = \frac{1}{2}mr^2$. The mass center of the disk is located at a distance of 0.25 m from point O . Thus,

$$\begin{aligned}m_d &= \rho_d V_d = 8000 \text{ kg/m}^3 [\pi(0.25 \text{ m})^2(0.01 \text{ m})] = 15.71 \text{ kg} \\(I_d)_O &= \frac{1}{2}m_d r_d^2 + m_d d^2 \\&= \frac{1}{2}(15.71 \text{ kg})(0.25 \text{ m})^2 + (15.71 \text{ kg})(0.25 \text{ m})^2 \\&= 1.473 \text{ kg} \cdot \text{m}^2\end{aligned}$$

Hole. For the 125-mm-radius disk (hole), we have

$$\begin{aligned}m_h &= \rho_h V_h = 8000 \text{ kg/m}^3 [\pi(0.125 \text{ m})^2(0.01 \text{ m})] = 3.927 \text{ kg} \\(I_h)_O &= \frac{1}{2}m_h r_h^2 + m_h d^2 \\&= \frac{1}{2}(3.927 \text{ kg})(0.125 \text{ m})^2 + (3.927 \text{ kg})(0.25 \text{ m})^2 \\&= 0.276 \text{ kg} \cdot \text{m}^2\end{aligned}$$

The moment of inertia of the plate about point O is therefore

$$\begin{aligned}I_O &= (I_d)_O - (I_h)_O \\&= 1.473 \text{ kg} \cdot \text{m}^2 - 0.276 \text{ kg} \cdot \text{m}^2 \\&= 1.20 \text{ kg} \cdot \text{m}^2\end{aligned}$$

Ans.

Example 6.4

The pendulum in Fig. E-6.4 is suspended from the pin at O and consists of two thin rods, each having a weight of 10 lb. Determine the moment of inertia of the pendulum about an axis passing through (a) point O , and (b) the mass center G of the pendulum.

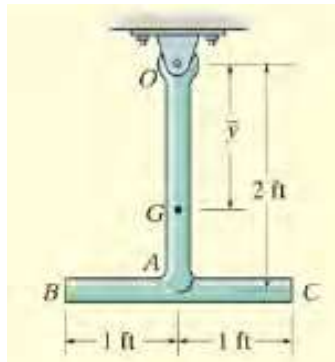


Fig. E-6.4

SOLUTION

Part (a). Using the table on the inside back cover, the moment of inertia of rod OA about an axis perpendicular to the page and passing through point O of the rod is $I_O = \frac{1}{3}ml^2$. Hence,

$$(I_{OA})_O = \frac{1}{3}ml^2 = \frac{1}{3}\left(\frac{10 \text{ lb}}{32.2 \text{ ft/s}^2}\right)(2 \text{ ft})^2 = 0.414 \text{ slug} \cdot \text{ft}^2$$

This same value can be obtained using $I_G = \frac{1}{12}ml^2$ and the parallel-axis theorem.

$$\begin{aligned}(I_{OA})_O &= \frac{1}{12}ml^2 + md^2 = \frac{1}{12}\left(\frac{10 \text{ lb}}{32.2 \text{ ft/s}^2}\right)(2 \text{ ft})^2 + \left(\frac{10 \text{ lb}}{32.2 \text{ ft/s}^2}\right)(1 \text{ ft})^2 \\ &= 0.414 \text{ slug} \cdot \text{ft}^2\end{aligned}$$

For rod BC we have

$$\begin{aligned}(I_{BC})_O &= \frac{1}{12}ml^2 + md^2 = \frac{1}{12}\left(\frac{10 \text{ lb}}{32.2 \text{ ft/s}^2}\right)(2 \text{ ft})^2 + \left(\frac{10 \text{ lb}}{32.2 \text{ ft/s}^2}\right)(2 \text{ ft})^2 \\ &= 1.346 \text{ slug} \cdot \text{ft}^2\end{aligned}$$

The moment of inertia of the pendulum about O is therefore

$$I_O = 0.414 + 1.346 = 1.76 \text{ slug} \cdot \text{ft}^2 \quad \text{Ans.}$$

Part (b). The mass center G will be located relative to point O . Assuming this distance to be $\bar{y}$, Fig. 17-7, and using the formula for determining the mass center, we have

$$\bar{y} = \frac{\sum \tilde{y}m}{\sum m} = \frac{1(10/32.2) + 2(10/32.2)}{(10/32.2) + (10/32.2)} = 1.50 \text{ ft}$$

The moment of inertia I_G may be found in the same manner as I_O , which requires successive applications of the parallel-axis theorem to transfer the moments of inertia of rods OA and BC to G . A more direct solution, however, involves using the result for I_O , i.e.,

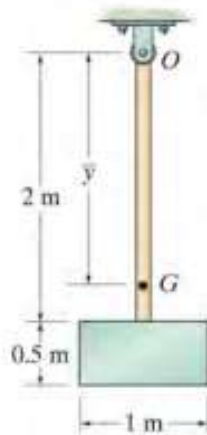
$$I_O = I_G + md^2; \quad 1.76 \text{ slug} \cdot \text{ft}^2 = I_G + \left(\frac{20 \text{ lb}}{32.2 \text{ ft/s}^2} \right) (1.50 \text{ ft})^2$$

$$I_G = 0.362 \text{ slug} \cdot \text{ft}^2$$

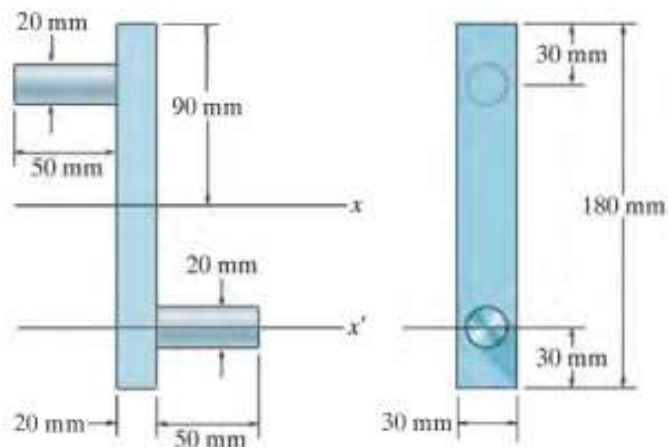
Ans.

Problem Set 6.1

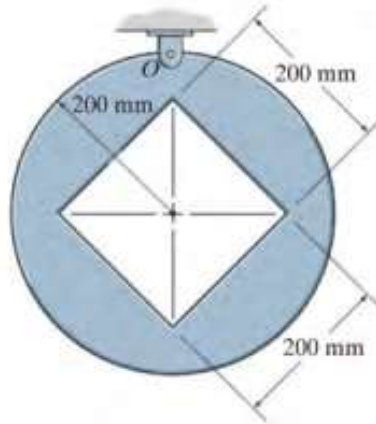
1. The pendulum consists of the 3-kg slender rod and the 5-kg thin plate. Determine the location $\bar{y}$ of the center of mass G of the pendulum; then calculate the moment of inertia of the pendulum about an axis perpendicular to the page and passing through G .



2. Determine the moment of inertia of the overhung crank (a) about the x axis and (b) about the x' axis. The material is steel for which the density is $\rho = 7.85 \text{ Mg/m}^3$.



3. Determine the mass moment of inertia of the thin plate about an axis perpendicular to the page and passing through point O. The material has a mass per unit area of 20 kg/m^2 .



6.2 Planar Kinetic Equations of Motion

In the following analysis we will limit our study of planar kinetics to rigid bodies which, along with their loadings, are considered to be symmetrical with respect to a fixed reference plane. Since the motion of the body can be viewed within the reference plane, all the forces (and couple moments) acting on the body can then be projected onto the plane.

6.2.1 Equation of translational motion

$$\sum \mathbf{F} = m\mathbf{a}_G \quad (6.5)$$

This equation is referred to as the **translational equation of motion** for the mass center of a rigid body. It states that **the sum of all the external forces acting on the body is equal to the body's mass times the acceleration of its mass center G .**

For motion of the body in the x-y plane, the translational equation of motion may be written in the form of two independent scalar equations, namely

$$\sum F_x = m(a_G)_x \quad (6.6)$$

$$\sum F_y = m(a_G)_y \quad (6.7)$$

Rectilinear translation: When a body is subjected to rectilinear translation, all the particles of the body (slab) travel along parallel straight line paths. Eqs 6.6 and 6.7 is applicable for *rectilinear translation*. In addition, the sum of all moments about the center of gravity is zero.

$$\sum M_G = 0 \quad (6.8)$$

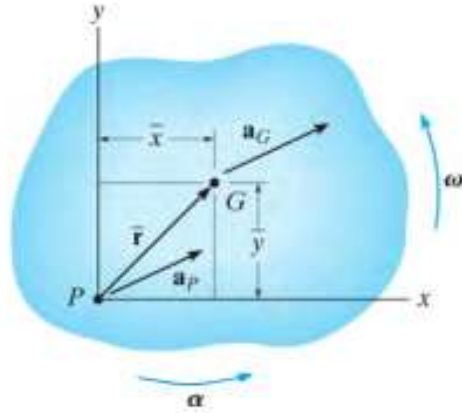
Curvilinear translation: When a rigid body is subjected to curvilinear translation, all the particles of the body travel along parallel curved paths. And the motion can be represented by

$$\sum F_n = m(a_G)_n \quad (6.9)$$

$$\sum F_t = m(a_G)_t \quad (6.10)$$

$$\sum M_G = 0 \quad (6.11)$$

6.2.2 Equation of rotational motion



We will now determine the effects caused by the moments of the external force system computed about an axis perpendicular to the plane of motion (the z axis) and passing through point P. The moment of the external forces acting on the body about point P can be expressed as

$$(+ACW) \sum M_P = -\bar{y}m(a_P)_x + \bar{x}m(a_P)_y + I_P\alpha \quad (6.8)$$

Where, $I_P = \int r^2 dm$

It is possible to reduce this equation to a simpler form if point P coincides with the mass center G for the body. If this is the case, then $\bar{x} = \bar{y} = 0$, and therefore

$$\sum M_G = I_G\alpha \quad (6.9)$$

This rotational equation of motion states that the sum of the moments of all the external forces about the body's mass center G is equal to the product of the moment of inertia of the body about an axis passing through G and the body's angular acceleration.

Example 6.5

The car shown in Fig. 17–10a has a mass of 2 Mg and a center of mass at G . Determine the acceleration if the rear “driving” wheels are always slipping, whereas the front wheels are free to rotate. Neglect the mass of the wheels. The coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$.

SOLUTION I

Free-Body Diagram. As shown in Fig. 17–10b, the rear-wheel frictional force $\mathbf{F}_B$ pushes the car forward, and since *slipping occurs*, $F_B = 0.25N_B$. The frictional forces acting on the *front wheels* are *zero*, since these wheels have negligible mass.* There are three unknowns in the problem, N_A , N_B , and a_G . Here we will sum moments about the mass center. The car (point G) accelerates to the left, i.e., in the negative x direction, Fig. 17–10b.

Equations of Motion.

$$\rightarrow \Sigma F_x = m(a_G)_x; \quad -0.25N_B = -(2000\text{ kg})a_G \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad N_A + N_B - 2000(9.81)\text{ N} = 0 \quad (2)$$

$$\zeta + \Sigma M_G = 0; \quad -N_A(1.25\text{ m}) - 0.25N_B(0.3\text{ m}) + N_B(0.75\text{ m}) = 0 \quad (3)$$

Solving,

$$a_G = 1.59\text{ m/s}^2 \leftarrow$$

Ans.

$$N_A = 6.88\text{ kN}$$

$$N_B = 12.7\text{ kN}$$

SOLUTION II

Free-Body and Kinetic Diagrams. If the “moment” equation is applied about point A , then the unknown N_A will be eliminated from the equation. To “visualize” the moment of $m\mathbf{a}_G$ about A , we will include the kinetic diagram as part of the analysis, Fig. 17–10c.

Equation of Motion.

$$\zeta + \Sigma M_A = \Sigma (\mathcal{M}_k)_A; \quad N_B(2\text{ m}) - [2000(9.81)\text{ N}](1.25\text{ m}) = (2000\text{ kg})a_G(0.3\text{ m})$$

Solving this and Eq. 1 for a_G leads to a simpler solution than that obtained from Eqs. 1 to 3.

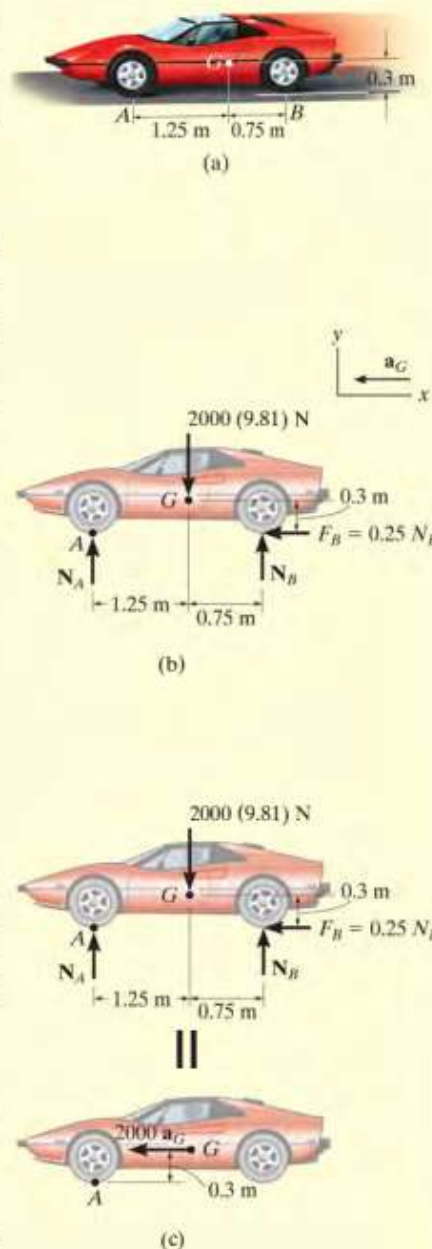


Fig. 17–10

Example 6.6



The motorcycle shown in Fig. 17-11a has a mass of 125 kg and a center of mass at G_1 , while the rider has a mass of 75 kg and a center of mass at G_2 . Determine the minimum coefficient of static friction between the wheels and the pavement in order for the rider to do a “wheelie,” i.e., lift the front wheel off the ground as shown in the photo. What acceleration is necessary to do this? Neglect the mass of the wheels and assume that the front wheel is free to roll.

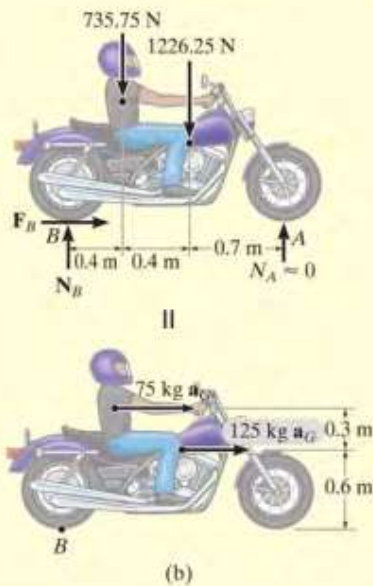


Fig. 17-11

SOLUTION

Free-Body and Kinetic Diagrams. In this problem we will consider both the motorcycle and the rider as a single *system*. It is possible first to determine the location of the center of mass for this “system” by using the equations $\bar{x} = \Sigma \tilde{x}m / \Sigma m$ and $\bar{y} = \Sigma \tilde{y}m / \Sigma m$. Here, however, we will consider the weight and mass of the motorcycle and rider separate as shown on the free-body and kinetic diagrams, Fig. 17-11b. Both of these parts move with the *same* acceleration. We have assumed that the front wheel is *about* to leave the ground, so that the normal reaction $N_A \approx 0$. The three unknowns in the problem are N_B , F_B , and a_G .

Equations of Motion.

$$\rightarrow \Sigma F_x = m(a_G)_x; \quad F_B = (75 \text{ kg} + 125 \text{ kg})a_G \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad N_B - 735.75 \text{ N} - 1226.25 \text{ N} = 0$$

$$\zeta + \Sigma M_B = \Sigma (\mathcal{M}_k)_B; \quad -(735.75 \text{ N})(0.4 \text{ m}) - (1226.25 \text{ N})(0.8 \text{ m}) = -(75 \text{ kg } a_G)(0.9 \text{ m}) - (125 \text{ kg } a_G)(0.6 \text{ m}) \quad (2)$$

Solving,

$$a_G = 8.95 \text{ m/s}^2 \rightarrow \quad \text{Ans.}$$

$$N_B = 1962 \text{ N}$$

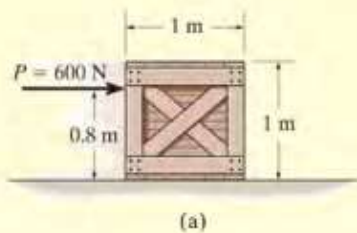
$$F_B = 1790 \text{ N}$$

Thus the minimum coefficient of static friction is

$$(\mu_s)_{\min} = \frac{F_B}{N_B} = \frac{1790 \text{ N}}{1962 \text{ N}} = 0.912 \quad \text{Ans.}$$

Example 6.7

A uniform 50-kg crate rests on a horizontal surface for which the coefficient of kinetic friction is $\mu_k = 0.2$. Determine the acceleration if a force of $P = 600$ N is applied to the crate as shown in Fig. 17-12a.



SOLUTION

Free-Body Diagram. The force $\mathbf{P}$ can cause the crate either to slide or to tip over. As shown in Fig. 17-12b, it is assumed that the crate slides, so that $F = \mu_k N_C = 0.2 N_C$. Also, the resultant normal force $\mathbf{N}_C$ acts at O , a distance x (where $0 < x \leq 0.5$ m) from the crate's center line.* The three unknowns are N_C , x , and a_G .

Equations of Motion.

$$\rightarrow \Sigma F_x = m(a_G)_x; \quad 600 \text{ N} - 0.2 N_C = (50 \text{ kg})a_G \quad (1)$$

$$+\uparrow \Sigma F_y = m(a_G)_y; \quad N_C - 490.5 \text{ N} = 0 \quad (2)$$

$$\zeta + \Sigma M_G = 0; \quad -600 \text{ N}(0.3 \text{ m}) + N_C(x) - 0.2 N_C(0.5 \text{ m}) = 0 \quad (3)$$

Solving,

$$N_C = 490.5 \text{ N}$$

$$x = 0.467 \text{ m}$$

$$a_G = 10.0 \text{ m/s}^2 \rightarrow \quad \text{Ans.}$$

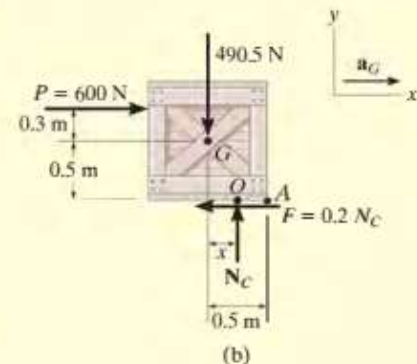
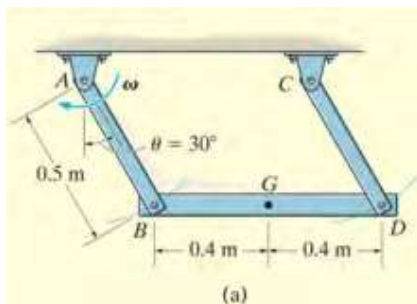


Fig. 17-12

Since $x = 0.467 \text{ m} < 0.5 \text{ m}$, indeed the crate slides as originally assumed.

NOTE: If the solution had given a value of $x > 0.5 \text{ m}$, the problem would have to be reworked since tipping occurs. If this were the case, $\mathbf{N}_C$ would act at the corner point A and $F \leq 0.2 N_C$.

Example 6.8



The 100-kg beam BD shown in Fig. 17-13a is supported by two rods having negligible mass. Determine the force developed in each rod if at the instant $\theta = 30^\circ$, $\omega = 6 \text{ rad/s}$.

SOLUTION

Free-Body Diagram. The beam moves with *curvilinear translation* since all points on the beam move along circular paths, each path having the same radius of 0.5 m. Using normal and tangential coordinates, the free-body diagram for the beam is shown in Fig. 17-13b. Because of the *translation*, G has the *same* motion as the pin at B , which is connected to both the rod and the beam. Note that the tangential component of acceleration acts downward to the left due to the clockwise direction of α , Fig. 17-13c. Furthermore, the normal component of acceleration is *always* directed toward the center of curvature (toward point A for rod AB). Since the angular velocity of AB is 6 rad/s when $\theta = 30^\circ$, then

$$(a_G)_n = \omega^2 r = (6 \text{ rad/s})^2 (0.5 \text{ m}) = 18 \text{ m/s}^2$$

The three unknowns are T_B , T_D , and $(a_G)_t$. The directions of $(a_G)_n$ and $(a_G)_t$ have been established, and are indicated on the coordinate axes.

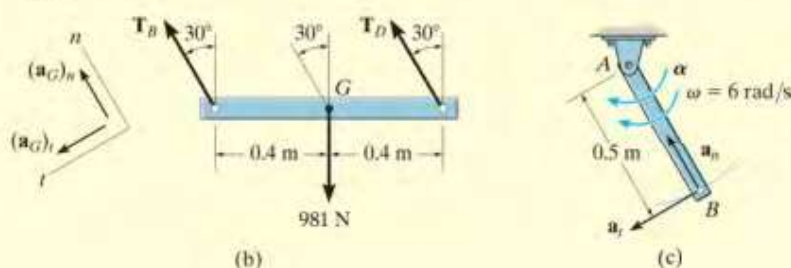


Fig. 17-13

Equations of Motion.

$$+\nearrow \Sigma F_n = m(a_G)_n; T_B + T_D - 981 \cos 30^\circ \text{ N} = 100 \text{ kg}(18 \text{ m/s}^2) \quad (1)$$

$$+\swarrow \Sigma F_t = m(a_G)_t; 981 \sin 30^\circ = 100 \text{ kg}(a_G)_t \quad (2)$$

$$\zeta + \Sigma M_G = 0; -(T_B \cos 30^\circ)(0.4 \text{ m}) + (T_D \cos 30^\circ)(0.4 \text{ m}) = 0 \quad (3)$$

Simultaneous solution of these three equations gives

$$\begin{aligned} T_B &= T_D = 1.32 \text{ kN} \\ (a_G)_t &= 4.905 \text{ m/s}^2 \end{aligned} \quad \text{Ans.}$$